GOVERNMENT OF WESTERN AUSTRALIA

# GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

*214315.1.pdf* *Geochronology Record 1907*

# 214315.1: metagranodiorite dyke, Wooley Spring

***(Big Bell Suite, Austin Downs Supersuite; Youanmi Terrane, Yilgarn Craton)***

**IOH Fielding, Y Lu, MTD Wingate and I Zibra**

## Location and sampling

YALGOO SH50-02, BADJA 2240 Zone 50, MGA 467528E 6825120N
Warox Site MZFYAD000121 Sampled on 21 May 2014

This sample was collected from low outcrop on the west side of a fenceline track on Bunnawarra Station, about 5.2 km east-northeast of New Bore, 4.3 km west-northwest of Brimson Well, and 0.6 km north of Wooley Spring.

## Geological context

The unit sampled is a metagranodiorite dyke assigned to the 2732–2686 Ma Big Bell Suite of the Austin Downs Supersuite (Van Kranendonk et al., 2013; Zibra, 2021: ENS Report Big Bell Suite, viewed 28 April 2023). The Big Bell Suite mainly consists of K-feldspar porphyritic and equigranular monzogranite and includes two north-trending batholiths exposed on either flank of the northeast-trending Meekatharra–Wydgee greenstone belt (Zibra, 2021: ENS Report Big Bell Suite, viewed 28 April 2023). At this locality, within the core of the Yalgoo dome, the dyke is north trending and crosscuts the foliation in the Kynea Migmatite of the Bunnawarra Suite (Fig. 1), a sample of which collected 1.5 km to the south yielded an igneous crystallization age of 2960 ± 10 Ma (GSWA 209689, Lu et al., 2017). The sample contains biotite-rich schlieren and inclusions of tonalite that are compositionally and microstructurally comparable to those from sample GSWA 209689 (Zibra et al., 2018). A metasyenogranite dyke of the Big Bell Suite sampled about 2.2 km to the south yielded an igneous crystallization age of 2705 ± 6 Ma (GSWA 214324, Lu et al., 2020).

**Figure 1. Outcrop image of sample 214315: metagranodiorite dyke, Wooley Spring. The sampled dyke with schlieren layering on the right crosscuts the foliation in a tonalite gneiss of the Kynea Migmatite**

## Petrographic description

The sample is a medium-grained metagranodiorite dyke (Fig. 1), consisting of about 40% plagioclase, 30% quartz, 15–20% K-feldspar, 5% biotite, 2–3% myrmekite, <1% opaques and accessory apatite and possible metamict allanite. Plagioclase is slightly sericitized, and mostly occurs as equant, anhedral grains up to 3 mm in size. Smaller xenoblastic plagioclase form polygonal and semipolygonal aggregates. Weakly strained quartz occurs as irregular fractured grains up to 4 mm in size.

The fine fractures appear to delineate subgrains and new grains <1 mm in size. Quartz forms small, rounded inclusions in larger plagioclase and K-feldspar grains. K-feldspar is finely microperthitic, equant, anhedral and up to 2.5 mm in size, and mostly exhibits fine microcline twinning. Myrmekite lobes up to 1 mm across occur on K-feldspar grain boundaries and within polygonal aggregates of smaller grains. Biotite occurs as randomly oriented, anhedral to subhedral laths up to 1.5 mm long, and is partly replaced by chlorite and trace muscovite. Probable magnetite, occurs as equant subhedral grains up to 0.2 mm in diameter. Allanite forms pale brown metamict tabular grains up to 0.4 mm long and apatite occurs as rare euhedral laths to 0.1 mm long. The sample lacks a tectonic foliation but contains evidence that it has been deformed and has undergone considerable solid-state grain boundary recovery at moderate metamorphic grade.

## Zircon morphology

Zircons isolated from this sample are colourless to dark brown and subhedral to euhedral. The crystals are up to 450 μm long and elongate with aspect ratios up to 6:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. Many crystals contain high uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

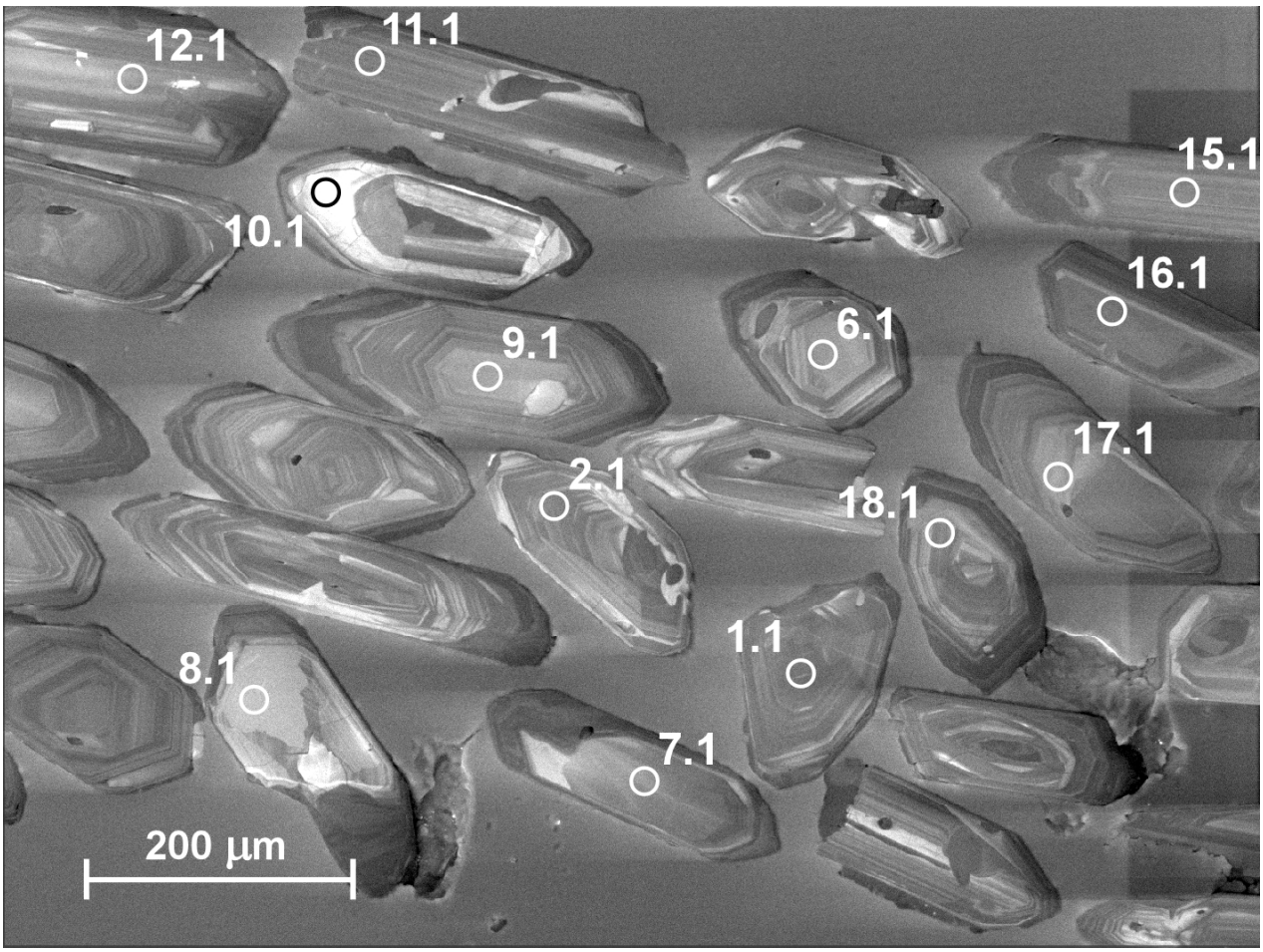


**Figure 2. Cathodoluminescence image of representative zircons from sample 214315: metagranodiorite dyke, Wooley Spring. Numbered circles indicate the approximate locations of analysis sites**

## Analytical details

This sample was analysed on 19–20 May 2015, using SHRIMP-B. Eleven analyses of the BR266 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.19% (1σ). Isotopic mass fractionation of $^{207}Pb^*/^{206}Pb^*$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.44%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

## Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

## Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 17 analyses of 17 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2700 ± 7 Ma (MSWD = 1.8).

Group X comprises a single analysis of a zircon core (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2958 ± 20 Ma (1σ).

The date of 2700 ± 7 Ma for the 17 analyses in Group I is interpreted as the magmatic crystallization age of the metagranodiorite dyke. The date of 2958 ± 20 Ma (1σ) for the analysis in Group X is interpreted as the age of an inherited component.

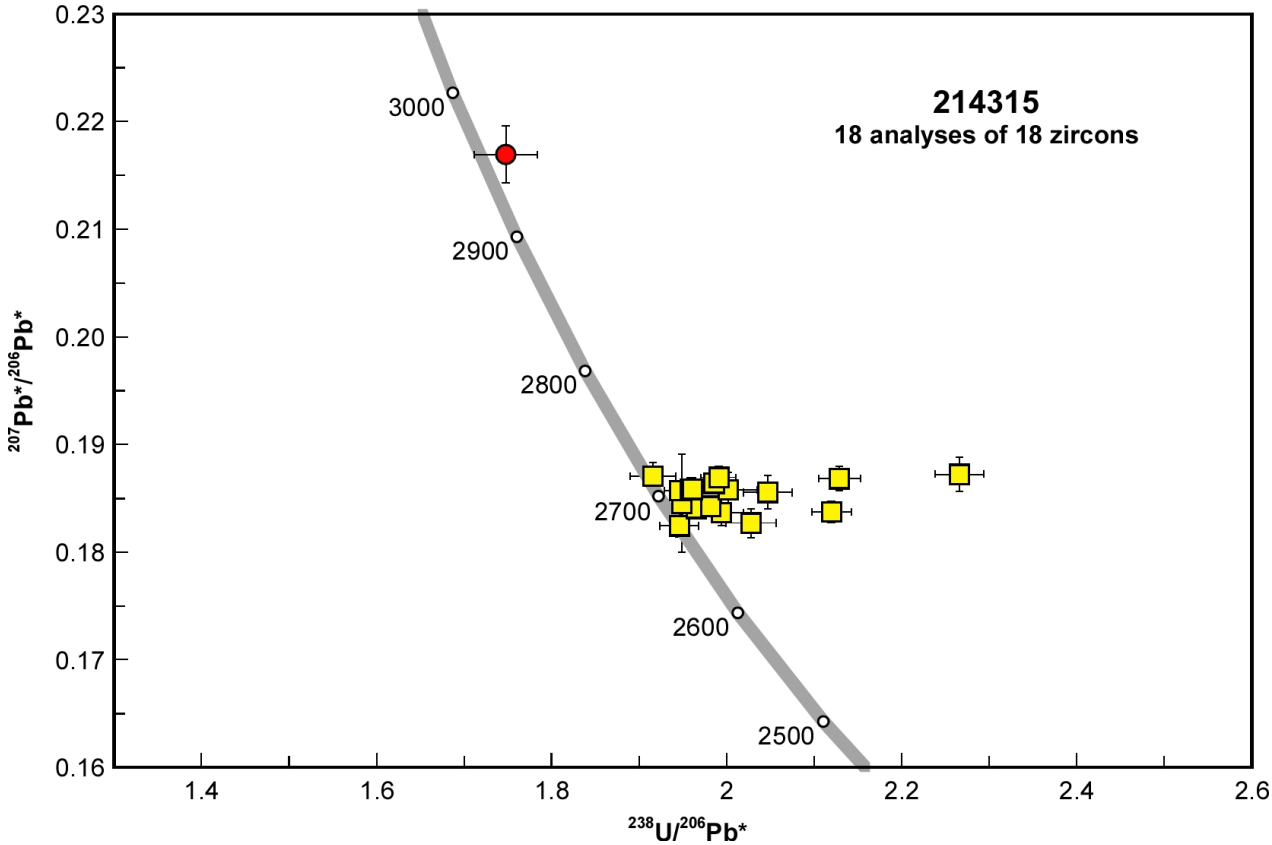


**Figure 3. U–Pb analytical data for zircons from sample 214315: metagranodiorite dyke, Wooley Spring. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon core)**

## Links

Table 1. Ion microprobe analytical results for zircons from sample 214315: metagranodiorite, Wooley Spring: 214315.1.Table1.xls
SHRIMP spot images: 214315_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

## References


Lu, Y, Wingate, MTD, Kirkland, CL and Zibra, I 2017, 209689: metatonalite, Wooley Spring; Geochronology Record 1463: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Lu, Y, Wingate, MTD and Zibra, I 2020, 214324: metasyenogranite dyke, Wooley Spring; Geochronology Record 1649: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Zibra, I, Peternell, M, Schiller, M, Wingate, MTD, Lu, YJ and Clos, F 2018, Tectono-magmatic evolution of the Neoarchean Yalgoo dome (Yilgarn Craton): diapirism in a pre-orogenic setting: Geological Survey of Western Australia, Report 176, 43p.

### *Recommended reference for this publication*

Fielding, IOH, Lu, Y, Wingate, MTD and Zibra, I 2023, 214315.1: metagranodiorite dyke, Wooley Spring; Geochronology Record 1907: Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract viewed 5 July 2023. <www.dmirs.wa.gov.au/geochron>

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

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